FINAL PROJECT • PART II • QUESTION 6

Paper Review

Mean estimation when you have the source code; or, quantum Monte Carlo methods

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Kothari and O'Donnell study how a quantum computer can estimate the mean of a random variable from the code generating its samples. Their algorithm achieves optimal query complexity [5, Theorem 18] without a supplied variance bound, removing logarithmic overhead from earlier general algorithms. Its new quantum core combines a value-dependent phase construction with spectral analysis.

Scope. I follow one complete route through the novel construction, correctness proof, and query analysis (Sections 3.1–3.4) to the general estimator (Section 4); Sections 1–2 supply the problem and access model. Within five pages, I use Section 3.5's sharper guarantee without its technical proof and omit the alternative Hadamard-test implementation (Section 3.6) and full spectral geometry (Section 3.7), neither needed for this route.

Problem, significance, and access model

For a real random variable y, write

$$\mu = \mathbb{E}[y], \qquad s^2 = \mathbb{E}[y^2], \qquad \sigma^2 = \mathrm{Var}(y) = s^2 - \mu^2. \tag{1}$$

Here s measures root-mean-square size relative to zero, whereas σ measures spread around the mean. Theorem 1.1 of Kothari and O'Donnell [1] gives a quantum algorithm using $O(n)$ invocations of the code for y and returning $\widehat{\mu}$ with

$$\Pr[|\widehat{\mu} - \mu| > \sigma/n] \le 1/3. \tag{2}$$

For comparison, the mean of n independent classical samples has standard deviation $\sigma/\sqrt{n}$. Thus constant-confidence error $\varepsilon\sigma$ costs $O(1/\varepsilon)$ quantum queries, versus $\Theta(1/\varepsilon^2)$ classical samples in general. Independent repetition and a median reduce failure to δ at an additional $O(\log(1/\delta))$ cost.

Amplitude estimation already improves on classical sampling for bounded quantities [2]; later work extends quantum summation and Monte Carlo [3, 4]. Hamoudi obtained the same variance-sensitive guarantee without a supplied variance bound, using $O(n \log^{3/2} n \log\log n)$ quantum experiments [5]. Kothari and O'Donnell remove these extra factors. They reuse only Hamoudi's quantile subroutine, costing $O(n)$ at the required tail probability, rather than his full mean estimator.

The model uses a finite probability space (Ω, p) with encoded real outputs $y(\omega)$. A preparation circuit P and reversible evaluator Y supply

$$P\left|\bar{0}\right\rangle = \sum_{\omega\in\Omega} \sqrt{p(\omega)}\,|\omega\rangle\,|g_\omega\rangle\,, \qquad Y : |\omega\rangle\,|0\rangle\,|0_{\mathrm{w}}\rangle \longmapsto |\omega\rangle\,|y(\omega)\rangle\,|0_{\mathrm{w}}\rangle\,. \tag{3}$$

The work register $|0_{\mathrm{w}}\rangle$ starts and ends in zero; P retains outcome-dependent, normalized workspace $|g_\omega\rangle$. The algorithm can apply controlled $P, P^\dagger, Y, Y^\dagger$ without measuring the outcome; each application is a code invocation. Given efficient classical sampling and evaluation code, reversible simulation constructs P and Y with constant-factor gate overhead: Hadamards supply the sampler's random bits, its workspace is retained, and evaluation uses compute–copy–uncompute (Section 2.1; Remark 2.11). Requiring garbage-free preparation would be substantially more restrictive. The guarantee requires these coherent operations, not merely measured samples.

The quantum construction

The paper first constructs a detector for a mean near zero; testing shifted versions of the variable will locate the mean numerically. This normalized Main Task (Theorem 1.3) is: given $\mathbb{E}[y^2] \le 1$ and a scale $\varepsilon > 0$, distinguish

$$|\mu| \le \varepsilon/2 \qquad \text{from} \qquad \varepsilon \le |\mu| \le 2\varepsilon \tag{4}$$

with success at least $2/3$ using $O(1/\varepsilon)$ queries.

The sharp phase bound requires $s \le 1/16$. Dividing y and ε by 16 scales μ, s likewise, preserves both promise cases, and changes the cost only to $O(1/(\varepsilon/16)) = O(1/\varepsilon)$. Through quantum phase estimation (QPE), these symbols denote the rescaled quantities.

The algorithm starts in the distribution state $P\left|\bar{0}\right\rangle$, denoted $|1\rangle$ in the paper: each amplitude has its original weight $\sqrt{p(\omega)}$, multiplied by the constant function 1. One iteration applies a phase depending on the outcome's value, then reflects about this starting state:

$$U = \mathrm{REFL}_p\,\mathrm{ROT}_y, \qquad \mathrm{REFL}_p = P(2\left|\bar{0}\right\rangle\left\langle\bar{0}\right| - I)P^\dagger = 2\,|1\rangle\langle 1| - I, \tag{5}$$

$$\mathrm{ROT}_y\,|\omega\rangle\,|g_\omega\rangle = e^{i\alpha_\omega}\,|\omega\rangle\,|g_\omega\rangle\,, \qquad \alpha_\omega = -2\arctan y(\omega). \tag{6}$$

These are the paper's equations (14)–(16). The phase rotates $1 + iy$ to $1 - iy$: they have equal length and opposite angles $\pm\arctan y$. The polar form makes the paper's equation (17) explicit, writing $y = y(\omega)$ pointwise:

$$e^{i\alpha_\omega}(1 + iy) = e^{-2i\arctan y}\sqrt{1+y^2}\,e^{i\arctan y} = \sqrt{1+y^2}\,e^{-i\arctan y} = 1 - iy. \tag{7}$$

To implement the phase operation, coherently compute $y(\omega)$ into a temporary register using Y, calculate α_ω, and apply its phase. Reverse the arithmetic and apply $Y^\dagger$: the temporary registers return to zero, but the phase remains on the outcome's amplitude. No outcome is measured.

For the reflection, apply $P^\dagger$, then $2\left|\bar{0}\right\rangle\left\langle\bar{0}\right| - I$, then P. This preserves the component along $|1\rangle$ and reverses every orthogonal component. One controlled iteration therefore uses controlled $Y, Y^\dagger, P^\dagger, P$: four code invocations, plus arithmetic and elementary gates.

In Grover search, marked and unmarked outcomes each evolve together, giving two collective directions. Here different values receive different phases, so many directions may participate. We must show that phase estimation usually returns a phase whose magnitude reveals $|\mu|$.

Why the construction reveals the mean

To track complex amplitude multipliers $z(\omega)$, the paper defines

$$|z\rangle = \sum_\omega \sqrt{p(\omega)}z(\omega)\,|\omega\rangle\,|g_\omega\rangle\,. \tag{8}$$

The target y remains real; $|z\rangle$ need not be normalized or prepared. For two such complex amplitude functions w, z, orthogonal outcome labels and normalized workspace give the paper's inner-product identity (Fact 3.10):

$$\langle w \mid z\rangle = \sum_\omega p(\omega)\overline{w(\omega)}z(\omega) = \mathbb{E}[\overline{w}z]. \tag{9}$$

Taking $w \equiv 1$ gives $\langle 1 \mid z\rangle = \mathbb{E}[z]$. To track what the reflection does to these multipliers, substitute this into its projection formula:

$$\mathrm{REFL}_p\,|z\rangle = (2\,|1\rangle\langle 1| - I)\,|z\rangle = 2\,|1\rangle\langle 1 \mid z\rangle - |z\rangle = 2\mathbb{E}[z]\,|1\rangle - |z\rangle = |2\mathbb{E}[z] - z\rangle\,. \tag{10}$$

Thus $z = \mathbb{E}[z] + (z - \mathbb{E}[z])$ becomes $\mathbb{E}[z] - (z - \mathbb{E}[z])$: each multiplier is reflected through the probability-weighted mean.

The paper chooses $z = 1 + iy$ because the phase identity sends it to $1 - iy$, whose mean is $1 - i\mu$. Reflecting about that mean produces the chain

$$|1 + iy\rangle \xrightarrow{\mathrm{ROT}_y} |1 - iy\rangle \xrightarrow{\mathrm{REFL}_p} |1 + i(y - 2\mu)\rangle = |1 + iy\rangle - 2i\mu\,|1\rangle\,. \tag{11}$$

The chain computes $U|1+iy\rangle$. Subtracting this image from $|1+iy\rangle$ cancels the outcome-dependent part and isolates the mean (Proposition 3.15):

$$\boxed{\frac{I-U}{2}|1+iy\rangle = \frac{|1+iy\rangle - U|1+iy\rangle}{2} = \frac{2i\mu|1\rangle}{2} = i\mu|1\rangle.} \tag{12}$$

There are now two states: the actual starting state $|1\rangle$, and the auxiliary proof state $\left|\widetilde{1}_y\right\rangle = |1+iy\rangle/\sqrt{1+s^2}$. The algorithm need not prepare the latter. The standard QPE rule explains why both are useful to analyze: for a normalized input $|\psi\rangle = \sum_j c_j \left|u_j\right\rangle$, where $U\left|u_j\right\rangle = e^{i\theta_j}\left|u_j\right\rangle$, ideal QPE returns θ_j with probability $|c_j|^2$. Hence an average $\mathbb{E}[f(\theta)]$ means $\sum_j |c_j|^2 f(\theta_j)$. Write θ for the ideal phase from $|1\rangle$ and $\widetilde{\theta}$ for that from $\left|\widetilde{1}_y\right\rangle$, using $-\pi < \theta_j \le \pi$. The auxiliary phase law is used only in the proof.

The vector identity becomes a statement about these averages by putting $M = (I-U)/2$: since $U\left|u_j\right\rangle = e^{i\theta_j}\left|u_j\right\rangle$, $M\left|u_j\right\rangle = \frac{1}{2}(\left|u_j\right\rangle - e^{i\theta_j}\left|u_j\right\rangle) = \frac{1-e^{i\theta_j}}{2}\left|u_j\right\rangle$. Orthogonality removes cross terms when taking the squared norm:

$$\|M|\psi\rangle\|^2 = \sum_j |c_j|^2 \left|\frac{1-e^{i\theta_j}}{2}\right|^2 = \sum_j |c_j|^2 \operatorname{hav}\theta_j = \mathbb{E}[\operatorname{hav}\theta], \qquad \operatorname{hav}\theta = \sin^2(\theta/2). \tag{13}$$

Normalizing (12) and taking squared norms therefore gives an average for the auxiliary state (Proposition 3.16):

$$M\left|\widetilde{1}_y\right\rangle = \frac{i\mu}{\sqrt{1+s^2}}|1\rangle \quad\Longrightarrow\quad \mathbb{E}[\operatorname{hav}\widetilde{\theta}] = \left\|M\left|\widetilde{1}_y\right\rangle\right\|^2 = \frac{\mu^2}{1+s^2}. \tag{14}$$

This average permits mostly tiny phases compensated by a few larger ones. But $1/\operatorname{hav}\theta$ is large when $|\theta|$ is tiny, so its average limits that possibility. To avoid division by zero, first assume $\mu \neq 0$ and no mass at $\theta = 0$; the paper perturbs U and takes a limit in the final phase bounds to cover excluded cases. Dividing $M|1+iy\rangle = i\mu|1\rangle$ by $i\mu$ and applying M^{-1} gives the first equality. The inverse has squared eigenvalue magnitudes $1/\operatorname{hav}\theta$, giving Proposition 3.17:

$$M^{-1}|1\rangle = \frac{1}{i\mu}|1+iy\rangle \implies \mathbb{E}\left[\frac{1}{\operatorname{hav}\theta}\right] = \left\|M^{-1}|1\rangle\right\|^2 = \frac{\||1+iy\rangle\|^2}{\mu^2} = \frac{\mathbb{E}[1+y^2]}{\mu^2} = \frac{1+s^2}{\mu^2}. \tag{15}$$

To bound a typical phase from both sides (Theorem 3.18), put $a = \mu^2/(1+s^2)$ and fix a large constant K. Markov bounds the probability of exceeding K times an average by $1/K$; apply it to the reciprocal for small phases and directly for large ones: $\operatorname{hav}\theta < a/K$ iff $1/\operatorname{hav}\theta > K/a$. Thus

$$\Pr[\operatorname{hav}\theta < a/K] \le 1/K, \qquad \Pr[\operatorname{hav}\widetilde{\theta} > Ka] \le 1/K. \tag{16}$$

The lower bound concerns the actual distribution, the upper bound the auxiliary one. Their overlap is $|\langle 1 \mid \widetilde{1}_y\rangle| = \sqrt{(1+\mu^2)/(1+s^2)}$, with $\mu^2 \le s^2$. Small s makes the states close, so their probabilities for any eigenphase event differ by at most s (proof of Theorem 3.18). Transferring the upper bound and taking a union bound gives $a/K \le \operatorname{hav}\theta \le Ka$ with probability at least $1 - 2/K - s$. This compares distributions, not individual outcomes θ and $\widetilde{\theta}$. Since $|\theta|/\pi \le |\sin(\theta/2)| \le |\theta|/2$ for $|\theta| \le \pi$, this encodes the mean magnitude in phases seen from the actual starting state:

$$\boxed{|\theta| = \Theta(|\mu|) \quad \text{with high probability for sufficiently small fixed } s.} \tag{17}$$

From spectral information to the QPE decision

The spectral argument relates the mean to an *ideal eigenphase* (Theorem 3.18); it has not measured the phase or estimated the mean. The weak bounds allow the same phase range for $|\mu| = \varepsilon/2$ and $|\mu| = \varepsilon$. Section 3.5 narrows the factors enough to separate these cases despite QPE error. Its sharper guarantee is used as a black box, omitting its proof as stated in the scope (Corollary 3.23): for $s \le 1/16$,

$$\Pr\left[\frac{4}{5}|\mu| \le \frac{|\theta|}{2} \le \frac{5}{4}|\mu|\right] \ge \frac{7}{9}. \tag{18}$$

QPE now measures an approximation to this phase. Run it on U, starting from $|1\rangle$, with additive phase accuracy $\varepsilon/6$ and failure at most $1/9$. For a superposition, its output can be viewed probabilistically as drawing an ideal θ with the eigenbasis weights and reporting a noisy θ'. With $\theta, \theta' \in (-\pi, \pi]$, QPE guarantees $\Pr[\big||\theta| - |\theta'|\big| > \varepsilon/6] \le 1/9$ (Remark 3.5).

The ideal phase violates the sharp spectral guarantee with probability at most $2/9$; QPE exceeds its chosen error with probability at most $1/9$. The union bound gives combined failure at most $2/9 + 1/9 = 1/3$. Otherwise,

$$\frac{4}{5}|\mu| - \frac{\varepsilon}{12} \le \frac{|\theta'|}{2} \le \frac{5}{4}|\mu| + \frac{\varepsilon}{12}. \tag{19}$$

The two promised cases are consequently separated:

$$|\mu| \le \varepsilon/2 \Longrightarrow |\theta'|/2 \le 17\varepsilon/24 < 0.71\varepsilon, \qquad |\mu| \ge \varepsilon \Longrightarrow |\theta'|/2 \ge 43\varepsilon/60 > 0.71\varepsilon. \tag{20}$$

Thresholding $|\theta'|/2$ at 0.71ε succeeds with probability at least $2/3$. QPE's controlled powers have total exponent $O(1/\varepsilon)$, giving the same bound on applications of U and code invocations. This proves the detector guarantee (Theorem 3.19), even without $|\mu| \le 2\varepsilon$. Dividing the original input and threshold by 16 recovers the Main Task at constant-factor cost (Theorem 1.3). The output remains one bit; Section 4 turns these decisions into numerical estimation with error σ/n.

Section 4: the reduction ladder

Section 4 uses the detector already proved for $s \le 1$, including its internal rescaling by 16. Problems 1–3 build the normalized estimator; Problems 4–5 provide scale-estimation tools; Problem 6 is a known-variance side result, deferred below; Problems 7–9 give the final generalization. Shifts, rescalings, and clipping require only constant original code invocations per transformed query. Constant amplification gives success at least $2/3$ throughout.

Problem 1: reuse the near-zero detector. The Main Task proved above distinguishes $|\mu| \le \varepsilon/2$ from $\varepsilon \le |\mu| \le 2\varepsilon$, assuming $s \le 1$, in $O(1/\varepsilon)$ queries. It returns one bit.

Problem 2: test proximity to a candidate. Still assuming $s \le 1$, the mean lies in $[-1, 1]$, although y need not. For a candidate b in this interval, set $z = (y - b)/2$. Since $\mathbb{E}[(y - b)^2] \le 2s^2 + 2b^2 \le 4$, its second moment is at most one. Its mean is $(\mu - b)/2$, so the zero detector with threshold $\varepsilon/2$ distinguishes $|\mu - b| \le \varepsilon/2$ from $\varepsilon \le |\mu - b| \le 2\varepsilon$, still in $O(1/\varepsilon)$ queries (Lemma 4.2).

Problem 3: turn decisions into a numerical estimate. This is robust binary search: since $|\mu| \le s \le 1$, start with $[-1, 1]$. Test the candidate one quarter along the current interval, at threshold half its width: retain the right half for "far" or the left three quarters for "near." Both contain the promise gap, so arbitrary answers there are safe. Repeating until width ε gives an additive-ε mean estimate (Lemma 4.3). Equal failure budgets across the logarithmically many stages cause the extra loglog cost. The trick is to buy higher confidence at cheap coarse stages, leaving constant confidence for expensive fine ones. For geometrically decreasing accuracies η_j ending at η_*, stage j costs $O(\eta_j^{-1} \log(1/\delta_j))$. Allocating $\delta_j = \exp[-C\sqrt{\eta_j/\eta_*}]$ keeps total failure bounded and total cost $O(1/\eta_*) = O(1/\varepsilon)$, without an extra $\log\log(1/\varepsilon)$ factor from chaining stages.

Problems 4–5: auxiliary scale-estimation tools. These $O(n)$-query black boxes will learn $s^2 = \mathbb{E}[y^2]$ without a supplied scale. Problem 4 estimates a Bernoulli mean to error σ/n (Lemma 4.4). Problem 5 converts $v \in [0, 1]$ to a Bernoulli outcome of conditional mean v, preserving $\mathbb{E}[v]$ and giving error $\sqrt{\mathbb{E}[v]}/n$ (Lemma 4.5). The latter will be applied to y^2; it does not yet give the original y's σ/n error.

The paper next proves the known-variance-bound variant, Problem 6. For dependency clarity, I defer its one-sample centering reduction to where it is reused, immediately before Problem 9.

The final route is now Problems 7–9: bounded-case error s/n, removal of boundedness and the lower-scale promise, then error σ/n, respectively.

Problem 7: learn the scale of a bounded signed variable. Assume $y \in [-1, 1]$ and $s \ge 1/n$, with s unknown. Learning s will permit normalization: Problem 5 applied to y^2 estimates s^2 to error $s/(2n) \le s^2/2$, after a constant budget increase. The lower-scale promise turns additive accuracy into a relative estimate; taking a square root gives $s/2 \le \widehat{s} \le 2s$. Thus $y/(2\widehat{s})$ has second moment at most one. Problem 3 with error $1/(4n)$, followed by rescaling, gives error at most s/n in $O(n)$ queries (Lemma 4.7).

Problem 8: remove the remaining promises by clipping. Any cap $B > 0$ makes the clipped-and-rescaled variable bounded, but a small cap risks mean distortion; a large cap risks violating the lower-scale promise.

Hamoudi's quantile routine balances these requirements in $O(n)$ queries using the square-root quantum speedup underlying search/minimum finding. This is its $O(\log(1/\delta)/\sqrt{p})$ quantile theorem at $p = \Theta(n^{-2})$ (with an appropriate constant) and constant confidence, not its full $O(n \log^{3/2} n \log\log n)$ mean estimator. With high probability,

$$\Pr(|y| \geq B) \geq n^{-2}, \qquad \Pr(|y| > B) \leq C_0 n^{-2}. \tag{21}$$

Let y_c be y clipped to $[-B, B]$. For $B > 0$, the event $|y| \geq B$ has probability at least $1/n^2$ and makes $(y_c/B)^2 = 1$. Hence $\mathbb{E}[(y_c/B)^2] \geq 1/n^2$: both promises of Problem 7 hold. Multiplying its estimate by B gives error $\sqrt{\mathbb{E}[y_c^2]}/n \leq s/n$, since clipping never increases magnitude. If $B = 0$, the clipped mean is zero. The second tail bound controls bias:

$$|\mathbb{E}[y] - \mathbb{E}[y_c]| \leq \mathbb{E}[|y| \mathbf{1}_{\{|y|>B\}}] \leq s\sqrt{\Pr(|y| > B)} = O(s/n). \tag{22}$$

Constant budget adjustments give error s/n in $O(n)$ queries, without a supplied range or second-moment bound (Lemma 4.8).

Centering reduction (paper's Problem 6). Since $s^2 = \sigma^2 + \mu^2$, a large mean can make s/n much larger than σ/n. Subtracting a typical sample removes this offset. For an independent sample m, Chebyshev gives $|m - \mu| \leq 2\sigma$ with probability at least $3/4$. Conditional on such m, $\mathbb{E}[(y - m)^2] = \sigma^2 + (\mu - m)^2 \leq 5\sigma^2$. This centering reduction converts variance control into second-moment control. With a supplied positive $\sigma_{\text{bound}} \geq \sigma$, divide by $\sqrt{5}\,\sigma_{\text{bound}}$ and use Problem 3; rescaling and adding m back gives error σ_{bound}/n in $O(n)$ queries (Lemma 4.6). A zero bound needs one sample.

Problem 9: obtain the unknown-variance guarantee. The final algorithm reuses centering but feeds $y - m$ into Problem 8 rather than Problem 3. Its second-moment scale is at most $\sqrt{5}\sigma$ on the centering event, and Problem 8 handles that scale without being told it: the supplied variance bound disappears. Amplify its success to failure $1/12$ and increase the budget by a constant factor. The resulting error is at most σ/n, with total failure $1/4 + 1/12 = 1/3$. Adding m back estimates the original mean in $O(n)$ queries without supplied range or variance bounds (Theorem 1.1).

Complexity, novelty, and assessment

Appendix A translates the $O(n)$ code invocations into gates and justifies finite precision. Under its floating-point assumptions, source gate cost S gives $O(nS) + O(n \log n (\log\log n)^2)$ CNOT and arbitrary one-qubit gates. After clipping and rescaling, $O(\log n)$-bit precision suffices; phase arithmetic and Bernoulli conversion account for the additive cost.

The phase encoding and its high-dimensional spectral analysis form the novel quantum core. Grover-type algorithms already used complex phases; the novelty is the mean-revealing construction and its analysis. QPE and Hamoudi's quantile routine are inherited tools. Most Section 4 reductions are classical, but preserving $O(n)$ throughout is essential.

Coherent access remains a substantive limitation: measured samples alone do not supply the required operations. Efficient code conversion does not establish practical depth or fault-tolerant cost. The paper settles the worst-case query complexity up to constant factors within its coherent-access model and has no formal open-problems section. I would further investigate concrete resource costs and weaker access models; these are directions suggested by this review. Finite numerical precision is already addressed in Appendix A.

References

[1] R. Kothari, R. O'Donnell. *Mean estimation when you have the source code*. SODA 2023, 1186–1215. Paper theorem, proposition, lemma, and equation numbers follow arXiv:2208.07544v1.
[2] G. Brassard et al. *Quantum amplitude amplification and estimation*. Contemp. Math. 305, 53–74 (2002).
[3] S. Heinrich. *Quantum summation with an application to integration*. Journal of Complexity 18(1), 1–50 (2002).
[4] A. Montanaro. *Quantum speedup of Monte Carlo methods*. Proceedings of the Royal Society A 471 (2015).
[5] Y. Hamoudi. *Quantum sub-Gaussian mean estimator*. ESA 2021, LIPIcs 204, Article 50.